

$$T(n) = aT(n/b) + \theta(n^k \log^p n)$$

where n = size of the problem

a = number of subproblems in the recursion and $a \geq 1$

n/b = size of each subproblem

$b > 1$, $k \geq 0$ and p is a real number.

Then,

Case 1: if $a > b^k$, then $T(n) = \theta(n^{\log_b a})$

Case 2: if $a = b^k$, then

(a) if $p > -1$, then $T(n) = \theta(n^{\log_b a} \log^{p+1} n)$

(b) if $p = -1$, then $T(n) = \theta(n^{\log_b a} \log \log n)$

(c) if $p < -1$, then $T(n) = \theta(n^{\log_b a})$

Case 3: if $a < b^k$, then

(a) if $p \geq 0$, then $T(n) = \theta(n^k \log^p n)$

(b) if $p < 0$, then $T(n) = \theta(n^k)$

Three parameters:

a : number of subproblems

b : factor by which input size shrinks

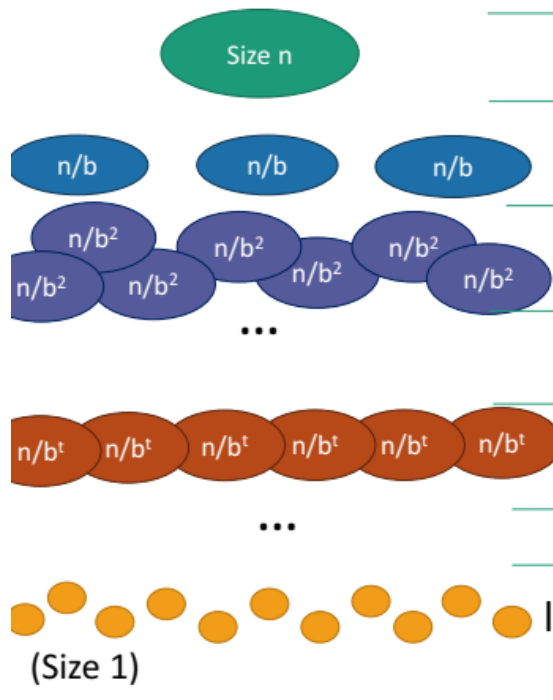
d : need to do nd work to create all the subproblems and combine their solutions.

$$T(n) = aT(n-b) + f(n)$$

- If $a=1$, $T(n) = O(n \cdot f(n))$
- If $a>1$, $T(n) = O(a^{n/b} \cdot f(n))$
- If $a<1$, $T(n) = O(f(n))$

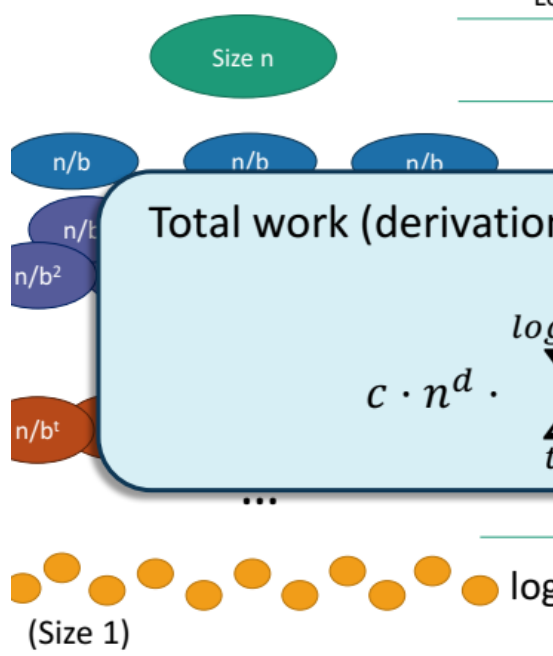
Recursion tree

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + c \cdot n^d$$

	Level	# problems	Size of each problem	Amount of work at this level
	0	1	n	$c \cdot n^d$
	1	a	n/b	$a c \left(\frac{n}{b}\right)^d$
	2	a^2	n/b^2	$a^2 c \left(\frac{n}{b^2}\right)^d$
...				
	t	a^t	n/b^t	$a^t c \left(\frac{n}{b^t}\right)^d$
...				
	$\log_b(n)$	$a^{\log_b(n)}$	1	$a^{\log_b(n)} c$
(Size 1)				

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Total work (derivation on board) is at most:

$$c \cdot n^d \cdot \sum_{t=0}^{\log_b(n)} \left(\frac{a}{b^d}\right)^t$$